# LOCO-Inspect AI: Multi-Category Industrial Visual Anomaly Detection

## Graduate Certificate in Pattern Recognition Systems — Project Proposal (First Presentation)

**Candidate:** Aryaman Srimali Jain | **Group Number:** 19

National University of Singapore • Institute of Systems Science (NUS-ISS)

Graduate Certificate in Pattern Recognition Systems — Practice Module

# 1. Industrial Motivation & Problem Statement



## The Automated Optical Inspection (AOI) Bottleneck

- High-speed modern manufacturing lines (semiconductors, bottling, consumer packaging) inspect thousands of components per hour.
- Traditional manual visual inspection is prone to fatigue, subjective bias, and high labor turnover.
- Supervised deep learning defect detectors fail fundamentally because real anomalies are rare, unpredictable, and cost-prohibitive to curate.
- Critical Need: An unsupervised pattern recognition system that models 'normality' exclusively from nominal, defect-free images.

## Core Engineering Objectives

- 100% Unsupervised Learning: Train purely on nominal reference images without viewing defect samples during training.
- Dual Anomaly Coverage: Capable of detecting both subtle physical surface defects and high-level logical constraint violations.
- Real-Time Throughput: Achieve sub-200ms GPU inference latency suitable for in-line conveyor deployment.
- Full-Stack Explainability: Provide operators with intuitive 4-panel visual localization overlays and bounding boxes.

# 2. The Benchmark: MVTec LOCO AD Dataset



## Dual Anomaly Paradigm

- MVTec Logical Constraints Anomaly Detection (LOCO AD) benchmark benchmarked by the machine vision industry.
- Structural Anomalies: Physical damages, scratches, cracks, cuts, material contamination, and surface deformities.
- Logical Anomalies: Products have nominal components, but violate logical assembly rules (e.g., missing items, count mismatches, wrong locations, inverted connectors).

## 5 Diverse Industrial Categories

- Juice Bottle: Liquid fill level violations, missing banana/orange fruit labels, physical bottle indentations.
- Breakfast Box: Granola, orange slices, almond count violations, foreign plastic wrap contamination.
- Pushpins Box: Excess/deficient pushpin counts, incorrect color assortment, broken transparent lid.
- Hardware Screw Bag: Missing M4 screws, incorrect nut quantities, torn polyethylene bags.
- Splicing Connectors: Loose copper wire terminals, misaligned latching clips, cracked plastic housing.

# 3. Proposed Deep Feature Extraction Pipeline



## WideResNet-50-2 Backbone Architecture

- Tapping intermediate convolutional activations rather than final classification heads.
- Layer 2 (Receptive Field ~40x40): Encodes fine texture, high-frequency edges, and surface micro-patterns.
- Layer 3 (Receptive Field ~80x80): Captures spatial layout, object part boundaries, and semantic context.
- Feature concatenation yields 1,024 localized spatial patch tokens of dimension 1,536 per image.

## Resolution & Invariance Properties

- Inputs standardized to 256x256 resolution with center-crop and ImageNet normalization.
- Permutation-aware representation retains spatial structural fidelity across nominal components.
- Zero backpropagation required during training — operates as a deterministic, reproducible feature encoder.

# 4. Algorithmic Strategy: PatchCore Memory Banks



## The Memory Scalability Challenge

- Raw patch memory bank accumulates 343,000+ normal vectors per category (~2.1 GB uncompressed RAM).
- Exhaustive k-Nearest Neighbor queries would result in intolerable multi-second latencies per inspection.
- Naive random subsampling degrades sensitivity to subtle defect boundaries.

## Johnson-Lindenstrauss & Coreset Selection

- Sparse Random Projection: Projects 1,536-D vectors down to 128-D while preserving pairwise Euclidean distances.
- Minimax Greedy Facility Location: Iteratively selects patches maximizing the minimum distance to the current subset.
- Achieves 90% memory compression (10% core memory bank) with zero loss in detection accuracy (0.896 ROC-AUC).

# 5. Parametric Statistical Modeling: PaDiM Baseline



## Location-Dependent Gaussian Formulation

- Per-Location Multivariate Gaussian: Fits a Gaussian distribution N(mu_{i,j}, Sigma_{i,j}) at each spatial patch coordinate (i,j).
- Estimated via Maximum Likelihood Estimation (MLE) across nominal training batches.
- Regularized Mahalanobis distance measures deviation from local coordinate-dependent normality.
- Directly tests whether coordinate-anchored statistics can resolve logical assembly rule violations.

## Comparative Evaluation Hypothesis

- Hypothesis 1: PatchCore memory banks will dominate on structural anomalies due to rich non-parametric coverage.
- Hypothesis 2: PaDiM will exhibit advantages on fixed-position logical constraints due to explicit spatial coordinate anchoring.
- Validation: Both algorithms will be comparatively benchmarked on identical test splits.

# 6. Inference Engine & High-Throughput Search



## Chunked GPU Nearest-Neighbor Search

- Custom TorchNN engine executes parallel distance computations on GPU tensor cores.
- Test image patches (1,024 vectors) queried against 34,304 coreset embeddings.
- Chunked tensor dot-products prevent GPU VRAM spikes and maintain sub-100ms response times.
- Calculates L2 Euclidean distance to nearest nominal prototype for every test patch.

## Anomaly Map Generation & Aggregation

- Spatial anomaly grid reshaped to 28x28 and upsampled via Bi-Cubic interpolation to 256x256.
- 2D Gaussian smoothing filter (sigma = 4) eliminates single-pixel sensor noise.
- Image-level anomaly score computed via adaptive top-k patch aggregation (capturing localized defects).

# 7. Explainable AI & Full-Stack Deployment Plan



## 4-View Visual Diagnostic HMI

- Panel 1 (Original Input): High-resolution RGB image uploaded by line operator.
- Panel 2 (Defect Bounding Box): Automated thresholding pinpoints peak defect coordinate rectangle.
- Panel 3 (Thermal Jet Heatmap): Color-coded anomaly intensity map (blue=normal, yellow=warning, red=defect).
- Panel 4 (Blended Overlay): Semi-transparent defect mask superimposed over the physical component.

## Full-Stack Software Architecture

- Backend: FastAPI asynchronous REST microservice (/api/inspect, /api/auth, /api/history).
- Security: SQLite database with PBKDF2-HMAC-SHA256 password hashing and audit trails.
- Frontend: Single-page application in HTML5/CSS3/JavaScript with real-time camera/sample ingestion.

# 8. Project Timeline, Work Breakdown & Milestones



## Phase 1: Foundation (Days 1 - 3)

- Literature review on unsupervised anomaly detection (SPADE, PaDiM, PatchCore).
- Dataset ingestion and exploratory analysis on MVTec LOCO AD categories.
- PyTorch feature extraction pipeline design with WideResNet-50-2.

## Phase 2: Modeling (Days 4 - 6)

- Memory bank construction & Minimax Facility Location Coreset algorithm implementation.
- PaDiM Gaussian MLE fitting and covariance regularizations.
- TorchNN GPU search optimization and threshold calibration (mu + k*sigma).

## Phase 3: Evaluation (Days 7 - 10)

- 5-category empirical benchmarking (ROC-AUC, Precision, Recall, F1, Pixel AUROC).
- Discovery and documentation of the 'Logical Anomaly Gap'.
- FastAPI web application development, testing, and final report compilation.

# 9. Target Success Criteria & Expected Deliverables



## Quantitative Performance Benchmarks

- Target Image ROC-AUC: Exceed 0.85 across all evaluation categories.
- Inference Speed: Sub-200ms per image on modern GPU; sub-1s on multi-core CPU.
- Zero Defect Escapes: High Sensitivity operating point with >95% recall for mission-critical production.

## Deliverable Package for Examination

- Runnable industrial AOI web application with interactive UI.
- Comprehensive 33-page technical report and reproducible Jupyter notebooks.
- 10-15 min video demonstration file and complete presentation decks.